Statistics in the turbulence talk figures

Formulas recovered from `codes/talk_figures.py`, `codes/paper_figures_li.py`, `turbulence/paper_figs_ensemble.py`, `codes/check_moments.py`.
Each formula is followed by its LaTeX code (copy from the grey box; math mode, no custom macros). Grey *Code:* lines are implementation details, not for the slides.

Notation (common to all figures)

Signals x of T samples $x(t)$, $t = 0, \dots, T-1$ (N signals). Lags τ are integers (samples). Lagrangian figures show the lag as τ/τ_η (τ_η: Kolmogorov time in samples); Eulerian (jets) figures show τ in samples.

$$\delta_\tau x(t) = x(t+\tau) - x(t)$$

```
\delta_\tau x(t) = x(t+\tau) - x(t)
```

$\mathbb{E}_x$: average over signals. $\mathbb{E}_t$: average over time. $\mathbb{E}_{x,t}$: both. Var, Std, Median, Quantile carry the same subscripts. All are empirical (over the signals shown).

Structure functions $S_p(\tau)$

Figures: `li_structure_functions` ($p = 2, 4, 6, 8$, Lagrangian); `fig6_*_1`, right panel ($p = 4, 6, 8$, both). Curves shifted down by 10^{-1} per order.

$$S_p(\tau) = \mathbb{E}_{x,t}\big[(\delta_\tau x)^p\big]$$

```
S_p(\tau) = \mathbb{E}_{x,t}\big[ (\delta_\tau x)^p \big]
```

Code: Fig. 6 takes the absolute value of the mean, $|\mathbb{E}_{x,t}[(\delta_\tau x)^p]|$: identical for even p. Lags: $\tau = 1, \dots, T/2-1$ (Fig. 6), $\tau = 1, \dots, \min(\texttt{max_lag}, T-2)$ (Li).

Flatness $F_p(\tau)$

Figure: `li_flatness` ($p = 4, 6, 8$, Lagrangian).

$$F_p(\tau) = \frac{S_p(\tau)}{S_2(\tau)^{p/2}}$$

```
F_p(\tau) = \frac{S_p(\tau)}{S_2(\tau)^{p/2}}
```

Gaussian values, for reference (not drawn in the figure):

$$F_4 = 3, \qquad F_6 = 15, \qquad F_8 = 105$$

```
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```

Flatness slope

Figure: `fig6_*_2`, left panel (both). Dotted line at 0.

$$\frac{d \log F_4}{d \log \tau}$$

```
\frac{d \log F_4}{d \log \tau}
```

Code: Evaluated on 30 log-spaced lags in $[1, T/2)$, centred second-order finite differences in $\log \tau$ (`np.gradient`).

Local scaling exponent $\zeta_p(\tau)$

Figure: `li_local_exponents` ($p = 4, 6, 8$, Lagrangian). Dotted line at $p/2$ (no intermittency: $S_p \propto S_2^{p/2}$).

$$\zeta_p(\tau) = \frac{d \log S_p}{d \log S_2}$$

```
\zeta_p(\tau) = \frac{d \log S_p}{d \log S_2}
```

Code: Computed as the ratio $\frac{d \log S_p / d \log \tau}{d \log S_2 / d \log \tau}$ (same thing, chain rule), finite differences on the unit-spaced lag grid $\tau = 1, 2, \ldots$, as Li et al. 2024.

Increment PDFs

$$\mathrm{PDF}\left(\frac{\delta_\tau x}{\mathrm{Std}_{x,t}[\delta_\tau x]} \right)$$

```
\mathrm{PDF}\!\left( \frac{\delta_\tau x}{\mathrm{Std}_{x,t}[\delta_\tau x]} \right)
```

`li_increment_pdfs` (Lagrangian): normalised density, $\tau/\tau_\eta = 1, 2, 5$, shifted down by 10^{-3} per lag. `fig6_*_2`, middle panel (both): PDF divided by its maximum (peak = 1), $\tau = 1, 4, 16, 64$ samples, shifted down by 10^{-1} per lag.

Code: Fig. 6: histogram on $[-6, 6]$, 201 bin edges, Gaussian smoothing of 1.5 bins. Li: no smoothing, bins over $\pm \max |\delta_\tau x| / \mathrm{Std}_{x,t}[\delta_\tau x]$.

Marginal PDF

Figure: `fig6_*_1`, left panel (both).

$$\mathrm{PDF}\left(\frac{x}{\mathrm{Std}_{x,t}[x]} \right)$$

```
\mathrm{PDF}\!\left( \frac{x}{\mathrm{Std}_{x,t}[x]} \right)
```

Code: The data are standardised ($\mathbb{E}_{x,t}[x] = 0$, $\mathrm{Std}_{x,t}[x] = 1$), so the code histograms x directly; the model samples are put on the same bins without re-standardising. Gaussian smoothing of 1.5 bins.

Power spectrum $P(\omega)$

Figure: `fig6_*_1`, middle panel (both); $\omega = k/T$ ($k = 1, \ldots, T/2$) in cycles per sample, shown for $0 < \omega \leq 0.4$.

$$P(\omega) = \mathbb{E}_x \Big[\Big| \sum_{t=0}^{T-1} x(t)\, e^{-2\pi i \omega t} \Big|^2 \Big]$$

```
P(\omega) = \mathbb{E}_x \Big[ \Big| \sum_{t=0}^{T-1} x(t)\, e^{-2\pi i \omega t}
    \Big|^2 \Big]
```

Code: No $1/T$ normalisation (`|rfft|`^{2} averaged over signals).

Skewness

Figure: `fig6_*_2`, right panel (both). Dotted line at 0.

$$\text{skewness}(y) = \mathbb{E}_{x,t}\left[\left(\frac{y - \mathbb{E}_{x,t}[y]}{\text{Std}_{x,t}[y]}\right)^3\right], \qquad y = x(t)\,\delta_\tau x(t)$$

```
\mathrm{skewness}(y) = \mathbb{E}_{x,t}\left[ \left( \frac{y -
    \mathbb{E}_{x,t}[y]}{\mathrm{Std}_{x,t}[y]} \right)^{3} \right], \qquad y =
    x(t)\,\delta_\tau x(t)
```

Code: `skew_mode='power'` (the talk setting). 45 log-spaced lags in $[1, T/2)$.

Cross-scale correlations $C_{p,q}(\tau, \tau^\star)$

Figure: `fig6_*_3` (both), $\tau^\star = 20$ samples (dotted vertical line; shown as $\tau^\star/\tau_\eta$ in the Lagrangian figure). Horizontal line at 1: value when the increment amplitudes at the two scales are uncorrelated.

$$C_{p,q}(\tau, \tau^\star) = \frac{\mathbb{E}_{x,t}\left[|\delta_{\tau^\star} x|^p\, |\delta_\tau x|^q\right]}{\mathbb{E}_{x,t}\left[|\delta_{\tau^\star} x|^p\right]\, \mathbb{E}_{x,t}\left[|\delta_\tau x|^q\right]}$$

```
C_{p,q}(\tau, \tau^\star) = \frac{\mathbb{E}_{x,t}\big[ |\delta_{\tau^\star} x|^p \,
    |\delta_\tau x|^q \big]}{\mathbb{E}_{x,t}\big[ |\delta_{\tau^\star} x|^p \big] \,
    \mathbb{E}_{x,t}\big[ |\delta_\tau x|^q \big]}
```

The three panels are $C_4 \equiv C_{2,2}$, $C_6 \equiv C_{3,3}$ and $C_{4,2}$. For even p, q the absolute values are not needed, e.g.

$$C_4(\tau, \tau^\star) = \frac{\mathbb{E}_{x,t}\left[(\delta_{\tau^\star} x)^2\, (\delta_\tau x)^2\right]}{S_2(\tau^\star)\, S_2(\tau)}$$

```
C_4(\tau, \tau^\star) = \frac{\mathbb{E}_{x,t}\big[ (\delta_{\tau^\star} x)^2 \,
    (\delta_\tau x)^2 \big]}{S_2(\tau^\star)\, S_2(\tau)}
```

Code: Both increments start at the same time t; the numerator averages over $t \le T - 1 - \max(\tau, \tau^\star)$, each denominator factor over its own full range. 10^{-8} added to the denominator. Lags $\tau = 1, \ldots, T/2 - 1$.

Rare signals: acceleration criterion

Acceleration = one-step increment (as Li et al.):

$$a(t) = \delta_1 x(t) = x(t+1) - x(t)$$

```
a(t) = \delta_1 x(t) = x(t+1) - x(t)
```

Each signal is summarised by its peak acceleration $\max_t |a(t)|$. **Criterion of the talk**: a signal is rare if

$$\max_t |a(t)| \;>\; \text{Quantile}_{0.95,\, x\in\text{data}}\left[\max_t |a(t)|\right]$$

```
\max_t |a(t)| \;>\; \mathrm{Quantile}_{0.95,\;x\in\mathrm{data}}\big[ \max_t |a(t)| \big]
```

i.e. rare = the 5 % of the data signals with the largest peak acceleration.

Alternative (`turb_rare_typical_sigma.ipynb`, threshold in units of the standard deviation of a): rare if

$$\max_t |a(t)| \;\geq\; 20\,\mathrm{Std}_{x,t\,;\,x\in\mathrm{data}}[a]$$

```
\max_t |a(t)| \;\ge\; 20\,\mathrm{Std}_{x,t\,;\,x\in\mathrm{data}}[a]
```

Code: One single number (every time of every data signal), so the threshold is the same level for the data and for the generated signals.

Code: `rare_events_data_vs_generated` applies the same rule at the lags $\tau/\tau_\eta = 20, 5, 1$, with a replaced by $\delta_\tau x$ and the quantile taken at that lag on the data; the generated signals use the same absolute threshold. `rare_events_by_logp` instead picks signals among the 1 % with the highest $-\log p$.

$-\log p$: what the mean and the std are taken over

The model is $p(x) \propto \exp(+\Theta \cdot \phi(x))$ ($\Theta = \Theta_{\mathrm{reg}}$ at the last time step), so

$$-\log p(x) = -\,\Theta \cdot \phi(x) + \mathrm{const}$$

```
-\log p(x) = -\,\Theta \cdot \phi(x) + \mathrm{const}
```

The constant is unknown, so the level of $-\log p$ means nothing: only differences between signals do. Large $-\log p$ = unlikely signal (rare signals sit on the right of the histograms). One fixed Θ; the $N = 10000$ data signals are the training signals:

$$\mathrm{mean} = \mathbb{E}_{x\in\mathrm{data}}\big[-\log p(x)\big], \qquad \mathrm{std} = \mathrm{Std}_{x\in\mathrm{data}}\big[-\log p(x)\big]$$

```
\mathrm{mean} = \mathbb{E}_{x\in\mathrm{data}}\big[ -\log p(x) \big], \qquad
    \mathrm{std} = \mathrm{Std}_{x\in\mathrm{data}}\big[ -\log p(x) \big]
```

- **Table** (per category c of potentials): the same $\mathbb{E}_{x\in\mathrm{data}}$ and $\mathrm{Std}_{x\in\mathrm{data}}$, applied to the contribution $-\sum_{F\in c} \Theta_F \cdot \phi_F(x)$ instead of $-\log p(x)$. Computed for each seed (Θ differs from seed to seed, the data signals are the same), then averaged over the seeds: $\mathbb{E}_{\mathrm{seed}}[\,\cdot\,]$.
- **Bar plots**: same quantities; error bars = $\mathrm{Std}_{\mathrm{seed}}$. Third panel: $\mathrm{Std}_x[\mathrm{category}]\,/\,\mathrm{Std}_x[-\log p]$.
- **Histograms** (data signals, or generated signals, each with its own median): $-\log p(x) - \mathrm{Median}_x\big[-\log p(x)\big]$.
- **Gaussian test**: $-\log p(x) - \mathbb{E}_{x\in\mathrm{train}}\big[-\log p(x)\big]$ for the exact $-\log p$, $-\log p(x) - \mathbb{E}_{x\in\mathrm{gen}}\big[-\log p(x)\big]$ for the MGD one. For M dimensions the exact $-\log p = \chi^2_M/2 + \mathrm{const}$, hence $\mathrm{Std} = \sqrt{M/2}$.

Code: $\mathbb{E}_{x\in\mathrm{data}}[-\log p]$ is a cross-entropy of the model on the data, not the entropy of the model (that would be the mean over samples of p). Code: `ddof=1` for every std ($N/(N-1) = 1.0001$ at $N = 10000$, so Std above is the same number to four digits). The sign flip leaves every std unchanged and changes the sign of every mean. Category contributions add up to $-\log p + \mathrm{const}$, but their stds do not add up (covariances between categories). The mean of a category contribution is a level only: a constant added to ϕ_F shifts it without changing p. `energy_table`, `plot_energy_bars`, `plot_logp_histogram`, `plot_logp_gaussian` in `codes/talk_figures.py`.

Error bands (all figures)

Line: the statistic on all signals. Band, `band='batch'` (default): min/max of the same statistic over 10 disjoint batches of signals (data and model, as Li et al.). `band='seeds'`: model line = mean over independent runs, band = ±1 std; the data band stays min/max over batches.